

3.8 Errors and Exceptions (Try-Except)

ایکسپشن (Exception) چلتے پروگرام کی غلطی ہے، جیسے غیر موجود فائل کھولنا۔

CS XII

Chapter 3: Programming Fundamentals

Lecture 16 of 18

3.8 Errors and Exceptions (Try-Except)

LEARN IT

An **exception** is an error raised while the program is running — for example opening a file that is not there. Without handling, Python prints a traceback and **stops**. A `try / except` block catches the error and lets you show a clear message instead of crashing.

Error type	Meaning
<code>FileNotFoundError</code>	The path does not exist.
<code>PermissionError</code>	The operating system will not let you read or write that file.
<code>IOError</code>	A general input/output problem (disk full, disk disconnected).

Put risky code in `try`. Put the recovery in `except ErrorType`. A final `except Exception as e` can catch anything else, using `e` as the error object.

```
try:
    with open('missing_data.txt', 'r') as f:
        content = f.read()
        print(content)
except FileNotFoundError:
    print("Error: The file does not exist. Please check the name.")
except Exception as e:
    print(f"An unexpected error occurred: {e}")
```

OUTPUT

Error: The file does not exist. Please check the name.

Without `try`, the same open raises:

OUTPUT

```
Traceback (most recent call last):
  File "missing.py", line 1, in <module>
    open('missing_data.txt', 'r')
FileNotFoundError: [Errno 2] No such file or directory: 'missing_data.txt'
```

WORKED EXAMPLE

When the file exists, the `except` blocks are skipped.

```
with open('ok.txt', 'w') as f:
    f.write("Present\n")

try:
    with open('ok.txt', 'r') as f:
        content = f.read()
        print(content, end="")
except FileNotFoundError:
    print("Error: The file does not exist. Please check the name.")
```

OUTPUT

Present

Professional software does not dump a red traceback on a user. If a file is missing or a disk fails, the user should see a calm sentence such as "Please check the file name" or "Please try again later."

REMEMBER IT

try = attempt; except = backup plan. Catch the **specific** error first (`FileNotFoundError`), then a general `Exception` if you need it. Order matters: a bare `except Exception` first would swallow the specific error.

COMMON MISTAKE

Do not wrap your entire program in one silent `except:` that ignores every mistake. You would hide real bugs. Catch the errors you expect, and keep the message honest. `'w'` and `'a'` create a missing file, so they do not raise `FileNotFoundError`; `'r'` does.

اردو میں سمجھیں

ایکسپشن (Exception) چلتے پروگرام کی غلطی ہے، جیسے غیر موجود فائل کھولنا۔ `try` میں خطرناک کوڈ لکھیں اور `except FileNotFoundError` میں صاف پیغام دکھائیں تاکہ پروگرام کریش نہ ہو۔ پیشہ ورانہ سافٹ ویئر صارف کو سرخ ٹریس بیک کی بجائے سمجھ آنے والا جملہ دکھاتا ہے۔

CHECK YOURSELF

1. Which exception is raised when `open('missing_data.txt', 'r')` cannot find the file?
2. Does the sample program crash, or print a message, when the file is missing?

Answers: 1. `FileNotFoundError`. 2. It prints "Error: The file does not exist. Please check the name." and continues.

Next lecture: 3.9 Mini File Handling Project: Grade Tracker